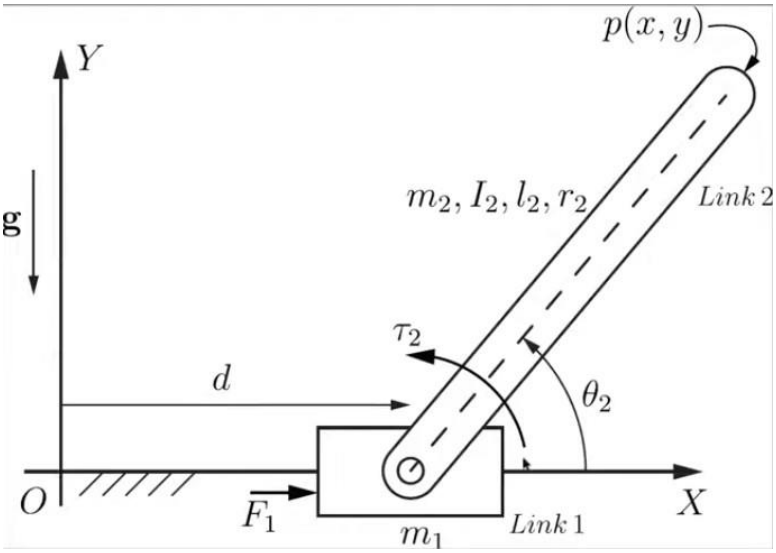
Mathematical Modelling



The cart-pole system is described by nonlinear equations of motion derived using Newtonian mechanics or the Lagrangian approach.

The model considers the interaction between the cart's horizontal movement and the pendulum's rotational movement.

The main physical parameters include:

- M: Mass of the cart (kg).
- m: Mass of the pendulum (kg).
- l: Distance from the pivot to the pendulum centre of mass (m).
- I: Pendulum moment of inertia about its centre of mass (kg·m²).
- g: Gravitational acceleration (m/s²).
- b: Cart viscous friction coefficient (N·s/m).
- F: Applied horizontal control force (N).

For an idealised system with cart friction represented by b, the nonlinear equations of motion can be expressed as:

Horizontal cart equation:

$$(M + m)\ddot{x} + b\dot{x} + ml\cos(\theta)\ddot{\theta} - ml\sin(\theta)\dot{\theta}^2 = F$$

Pendulum rotational equation:

$$(I + ml^2)\ddot{\theta} + ml\cos(\theta)\ddot{x} - mgl\sin(\theta) = 0$$

The exact signs depend on the chosen angular coordinate convention. In this project, the angle is defined relative to the upright vertical position.